Simulated annealing

Simulated annealing method was inspired by metallurgy, an alloy is heated and then cooled down, so its composition can reach a lower energy state. In programming the ultimate goal is the same – we want to find the best solution possible by simulating the process described above. The whole algorithm is controlled by its “temperature” - a float variable. At the beginning of every iteration we have a potential solution *S* and we consider its neighbours (*S* with slight modifications) in random order. If the considered solution *T* is better then *S*, we replace *S* with it, if it is worse – we replace S with *T* with a probability given by:

$$p = \exp\left(\frac{E_S - E_T}{t}\right)$$

where *E* is the value of the goal function we want to minimize and *t* is the temperature. By controlling *t*, we can decide whether we want to explore the solution space or just tend to the minimum.

1. Try to use the technique described above to find the global minimum of a polynomial $f(x) = x^6 - 7x^5 + 7x^4 + 35x^3 - 56x^2 - 28x + 48, \quad x \in (-2,4)$

2. Implement TSP using simulated annealing, neighbours of the solution *S* can be generating by switching different pairs of cities.

3. Use this approach to solve the Minimum Vertex Cover problem.